

SAREENA RAHIM

SOFTWARE DEVELOPER | PYTHON BACKEND

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PROFESSIONAL SUMMARY

Backend-focused software developer with a foundation in analytical problem-solving from a physics background and strong communication skills developed through prior mentoring work. Hands-on experience building event-driven backend systems in Python and FastAPI, including a fraud detection pipeline using Kafka, Redis, and PostgreSQL. Skilled in REST API design, authentication, and containerized development, with a fast-learning mindset and eagerness to grow in a backend or full-stack engineering role.

SKILLS

Programming Languages: Python, SQL, Java

Backend: FastAPI, REST API Design, Authentication & Authorization (JWT / token-based)

Frontend: HTML, CSS, JavaScript, React

Databases: PostgreSQL, MySQL, SQLite, Supabase

Tools & Infrastructure: Git, GitHub, Docker, Kafka, Redis

Concepts: Event-Driven Architecture, Transaction Processing, Database Design, Debugging & Troubleshooting

WORK EXPERIENCE

AMC Aerospace Technologies - UAE

Software Engineer Trainee

Nov 2025 – Jun 2026

- Enterprise Application Development:** Developed a currency module within a Java/JSF enterprise web application, and contributed to building and testing other features during hands-on development training.
- ERP Workflow Analysis:** Traced backend logic and data flow across ERP modules to understand end-to-end business workflows spanning multiple departments.
- UX Improvement & Debugging:** Identified usability problems in the existing user experience and resolved several of them through backend changes.
- Professional Adaptation:** Ramped up quickly on unfamiliar enterprise technologies and an established professional codebase.

Gangothri Community Development Project - Kerala

Teacher cum Counselor

Oct 2024 – Apr 2025

- Mentorship & Communication:** Mentored students in academic and personal development, strengthening their communication and leadership skills.
- Program Coordination:** Planned and executed educational and community programs in collaboration with a team.

KEY PROJECTS

Fraud Detection System | Python, FastAPI, PostgreSQL, Kafka, Redis, Docker

- Event-Driven Backend:** Built a fraud detection backend in Python/FastAPI that ingests transaction events via Kafka and flags suspicious activity against rule-based detection logic.
- Real-Time Velocity Checks:** Used Redis to maintain rolling transaction and velocity windows for real-time behavioral analysis.
- Data & Persistence:** Modeled transaction and user data in PostgreSQL for reliable storage and retrieval.
- Containerization:** Containerized supporting services with Docker for reproducible local development and testing.

Expense Tracker Application | FastAPI, Supabase, Tkinter

- Multi-User System:** Developed a multi-user expense tracker with token-based authentication and user-specific data isolation.
- REST API:** Built CRUD REST APIs with FastAPI for creating, reading, updating, and deleting expense records.
- Auth & Database:** Integrated Supabase for authentication and database management with secure API communication.
- Desktop UI:** Designed a Tkinter desktop interface for login and day-to-day expense tracking.

EDUCATION

Master of Social Work (MSW) — Nehru Arts and Science College, Coimbatore

2024

B.Sc. Physics — Sanatana Dharma College

2022